Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.
- My student hour: Tuesdays 1.00pm - 2.00pm, Bragg building foyer.

Revision

- If m and n are positive integers, there are unique integers q and r satisfying $m = qn + r$ and $0 \leq r < n$. We say q is the **quotient** and r is the **remainder** when we divide m by n.
- We write $a \equiv b \pmod{n}$ to mean that n divides $a - b$. This is an equivalence relation on integers. $5 \equiv 12 \pmod{7}$
- A **residue** modulo n is one of the numbers $\{0, 1, 2, \ldots, n-1\}$.
- We write $a \bmod n$ to represent the remainder (residue) we obtain when dividing a by n. Therefore $a \bmod n$ is always in $\{0, 1, 2, \ldots, n-1\}$. $15 \bmod 4 = 3$

  $17 \bmod 9 = 8$

Revision

Repeated squaring modulo n

Question. Calculate 2^{39} mod 9.

$39 = 32 + 4 + 2 + 1 \qquad (39)_{10} = (100111)_2$

i	y_i	z	p
		1	$2 \bmod 9$
0	1	$1 \times 2 \bmod 9 = 2$	$4 \bmod 9$
1	1	$2 \times 4 \bmod 9 = 8$	$7 \bmod 9$
2	1	$8 \times 7 \bmod 9 = 2$	$4 \bmod 9$
3	0	$2 \bmod 9$	$7 \bmod 9$
4	0	$2 \bmod 9$	$4 \bmod 9$
5	1	$2 \times 4 \bmod 9 = 8$	

square

square

($4^2 \bmod 9 = 16 \bmod 9 = (1 \times 9 + 7) \bmod 9 = 7$)

Euclidean algorithm

Let m and n be positive integers. If d divides both m and n, it is a **common divisor**. The **Euclidean algorithm** finds the **greatest common divisor** (written as $\gcd(m, n)$).

- Set r_0 to be m, and r_1 to be n.
- Repeatedly define $r_i = r_{i-2} \bmod r_{i-1}$.
- If r_t is the first zero entry in the sequence, then r_{t-1} is equal to $\gcd(m, n)$.

Find gcd(42, 30)

$r_0 = 42$

$r_1 = 30$

$r_2 = 42 \bmod 30 = 12$

$r_3 = 30 \bmod 12 = 6$

$r_4 = 12 \bmod 6 = 0$

$\gcd(42, 30) = 6$

Running time of the Euclidean algorithm

$\gcd(m,n) = \gcd(n,r)$

$m = qn + r \quad (0 \leq r < n)$

In general $\gcd(m,0) = m$

Proposition 9.15. Let m and n be non-negative integers such that $m \geq n$. If $r = m \bmod n$ then $r < m/2$.

Proposition 9.16. Let m and n be non-negative integers satisfying $m \geq n$. Let q be the smallest integer such that $2^q > m$. Define the sequence $r_0, r_1, r_2, \ldots$ by $r_0 = m$, $r_1 = n$, and $r_i = r_{i-2} \bmod r_{i-1}$. Then there is some integer $p \leq 2q$ such that $r_p = 0$.

q is the number of binary digits needed to represent m. The number of loops in the Euclidean algorithm is $\leq 2q$.

Euclidean algorithm

Input: Non-negative integers m and n where $m \geq n$.
Output: The greatest common divisor $r_{\text{old}} = \gcd(m, n)$.

$r_{\text{old}} \leftarrow m$
$r_{\text{new}} \leftarrow n$
while $r_{\text{new}} > 0$ **do**
    $t \leftarrow r_{\text{new}}$
    $r_{\text{new}} \leftarrow r_{\text{old}} \bmod t$
    $r_{\text{old}} \leftarrow t$
end while
return r_{old}

The extended Euclidean algorithm

Proposition 9.17. Let m and n be positive integers. There exist integers x and y such that $xm + yn = \gcd(m, n)$.

For paper-and-pen examples, we can find the integers x and y by keeping track of the quotients and remainders as we proceed through the Euclidean algorithm, and then backtracking through the equations, carrying out substitutions.

$\gcd(7, 4) = 1 = 2 \times 4 - 1 \times 7$

in this case $x7 + y4 = \gcd(7, 4)$ where $x = -1$, $y = 2$

The extended Euclidean algorithm

Example. Find $\gcd(62, 23)$ and integers x and y such that $\gcd(62, 23) = 62x + 23y$.

$62 = 2 \times 23 + 16$

$23 = 1 \times 16 + 7$

$16 = 2 \times 7 + 2$

$7 = 3 \times 2 + 1$

$2 = 2 \times 1 + 0$

$\gcd(62, 23) = 1$

$= 62x + 23y$

where $x = -10$, $y = 27$

$1 = 7 - 3 \times 2$

$= 7 - 3(16 - 2 \times 7)$

$= 1 \times 7 - 3 \times 16 + 6 \times 7$

$= 7 \times 7 - 3 \times 16$

$= 7(23 - 16) - 3 \times 16$

$= 7 \times 23 - 7 \times 16 - 3 \times 16$

$= 7 \times 23 - 10 \times 16$

$= 7 \times 23 - 10(62 - 2 \times 23)$

$= 7 \times 23 - 10 \times 62 + 20 \times 23$

$= 27 \times 23 - 10 \times 62$

Exam preparation

Example. Find $\gcd(33, 27)$ and integers x and y such that $\gcd(33, 27) = 33x + 27y$.

$33 = 1 \times 27 + 6$

$27 = 4 \times 6 + 3$

$6 = 2 \times 3 + 0$

$\gcd(33, 27) = 3$

$= 33x + 27y$

where $x = -4$, $y = 5$

$3 = 27 - 4 \times 6$

$= 27 - 4(33 - 27)$

$= 27 - 4 \times 33 + 4 \times 27$

$= 5 \times 27 - 4 \times 33$

The extended Euclidean algorithm

If we were implementing the extended Euclidean algorithm, we would use this algorithm, which uses temporary variables to store values, rather than storing the entire sequence of equations.

Input: Non-negative integers m and n where $m \geq n$.
Output: The greatest common divisor $r_{\text{old}} = \gcd(m, n)$ and integers x_{old} and y_{old} such that $\gcd(m, n) = x_{\text{old}}\, m + y_{\text{old}}\, n$.

$r_{\text{old}} \leftarrow m$
$r_{\text{new}} \leftarrow n$
$x_{\text{old}} \leftarrow 1$
$x_{\text{new}} \leftarrow 0$
$y_{\text{old}} \leftarrow 0$
$y_{\text{new}} \leftarrow 1$
while $r_{\text{new}} > 0$ **do**
    $t \leftarrow r_{\text{new}}$
    $(q, r) \leftarrow$ integers such that $r_{\text{old}} = qt + r$ and $0 \leq r < t$
    $r_{\text{new}} \leftarrow r$
    $r_{\text{old}} \leftarrow t$
    $t_x \leftarrow x_{\text{new}}$
    $x_{\text{new}} \leftarrow x_{\text{old}} - qt_x$
    $x_{\text{old}} \leftarrow t_x$
    $t_y \leftarrow y_{\text{new}}$
    $y_{\text{new}} \leftarrow y_{\text{old}} - qt_y$
    $y_{\text{old}} \leftarrow t_y$
end while
return $(r_{\text{old}}, x_{\text{old}}, y_{\text{old}})$

Multiplicative inverses

Let n be a positive integer, and let a and b be residues modulo n.

If $ab \bmod n = 1$, then we say that a and b are **inverses modulo n**.

If a has an inverse modulo n, then we write it as $a^{-1} \bmod n$.

What is $5^{-1} \bmod 11$?

$5 \times 9 \bmod 11$

$45 \bmod 11$

Because $45 = 4 \times 11 + 1$

we have

$45 \bmod 11 = 1$

$5^{-1} \bmod 11 = 9$

What is $2^{-1} \bmod 4$?

There is no inverse, because any multiple of 2 is even, so cannot have remainder 1 mod 4

Multiplicative inverses

If $\gcd(n, a) = 1$, then we say that n and a are **coprime**

Proposition 9.18. Let n be a positive integer and let a be a residue modulo n. Then a has a multiplicative inverse modulo n if and only if n and a are coprime.

Proof of one direction, the "only if" direction.
Assume that a has an inverse modulo n.
Call this inverse b. Therefore $ab \bmod n = 1$,
so $ab = qn + 1$ * for some quotient $q \in \mathbb{Z}$.
Let d be gcd(n, a). Since $d \mid a$, we have
$d \mid (ab)$. Since $d \mid n$, we have $d \mid (-qn)$.
So d divides $ab - qn = 1$ (by rearranging *).
d is a positive integer and $d \mid 1$, so $d = 1$. □

Finding multiplicative inverses

We can use the extended Euclidean algorithm to find the multiplicative inverse of a modulo n.

We assume a is coprime to n (or else the inverse does not exist). Therefore $\gcd(a, n) = 1$.

- Use the extended Euclidean algorithm to find integers x and y such that $xn + ya = 1$.
- Find the residue of y modulo n. This is the number in $0, 1, 2, \ldots, n-1$ that has the same remainder as y modulo n.
- This residue is the inverse of a.

Finding multiplicative inverses

Example. Find the inverse of 8 modulo 11.

$11 = 1 \times 8 + 3$

$8 = 2 \times 3 + 2$

$3 = 1 \times 2 + 1$

$2 = 2 \times 1 + 0$

$\gcd(11, 8) = 1$

$1 = 3 - 2$

$= 3 - (8 - 2 \times 3)$

$= 3 - 8 + 2 \times 3 = 3 \times 3 - 8$

$= 3(11 - 8) - 8$

$= 3 \times 11 - 3 \times 8 - 8$

$= 3 \times 11 - 4 \times 8$

$8^{-1} \bmod 11$ is the remainder of -4 mod 11.

$-4 = (-1) \times 11 + 7.$ So $8^{-1} \bmod 11 = 7$

Finding multiplicative inverses

Example. Find the inverse of 6 modulo 17.

Pseudorandom number generation

Random numbers are useful in many applications, especially in cryptography.

It is very hard to produce truly random numbers using a computer.

Instead, it is sometimes sufficient to produce a sequence that appears random. Such a sequence is called **pseudorandom**.

Pseudorandom number generation

- Choose n, a very large prime number.
- Choose integers a and c such that $0 < a, c < n$.
- Choose a seed integer x_0 such that $0 \leq x_0 < n$.
- Define x_{i+1} to be

$$x_{i+1} = \begin{cases} (ax_i^{-1} + c) \bmod n & \text{if } x_i \neq 0 \\ c & \text{if } x_i = 0 \end{cases}$$

If $n = 1013$, $a = 520$, $c = 617$, and $x_0 = 97$ the sequence starts 97, 873, 758, 754, 967, 914, 755, 81, 786, 878, 313, ...